

Through-life Engineering Services (TES): (#3) Service Capabilities (HOW?)

Author: Andrew Harrison (DFS Consulting Limited)

Sponsor: Bloc Digital/ATI/NATEP Published Date: 14 May 2024

Summary

Background

In this paper we expand upon the previous white paper discussions of Through Life Engineering Service (TES) value propositions¹, and service options², addressing the questions of how services practically delivered and what capabilities are required to deliver them?

The previous service options paper concluded that services (or their constituent elements) generally fall within the 4 value proposition themes of Usability, Reliability, Availability and Value which collectively provide a user/owner with the ability to effectively use, healthy and capable assets, that are in the right quantity, at the right place and the right time, whilst extracting the maximum value from their use, at the lowest direct and indirect cost. This paper will explore the underlying capability requirements to deliver this array of service elements.

Questions this paper addresses

- What activities does a Through-Life service need to deliver?
- What capabilities are required to enable the activities?

Conclusion

Through-Life Services are typically provided as a set of service lines or capabilities that address one or more of the Usability, Reliability, Availability or net asset Value themes.

The underlying capabilities, that enable these services to be provided, can broadly be split into 3 groups:

- the ability to deliver physical activity that impacts the physical asset (e.g. transport, maintain, inspect etc).
- the ability to decide what physical activity to undertake and when.
- the ability to contract, resource and financially manage the above activities.

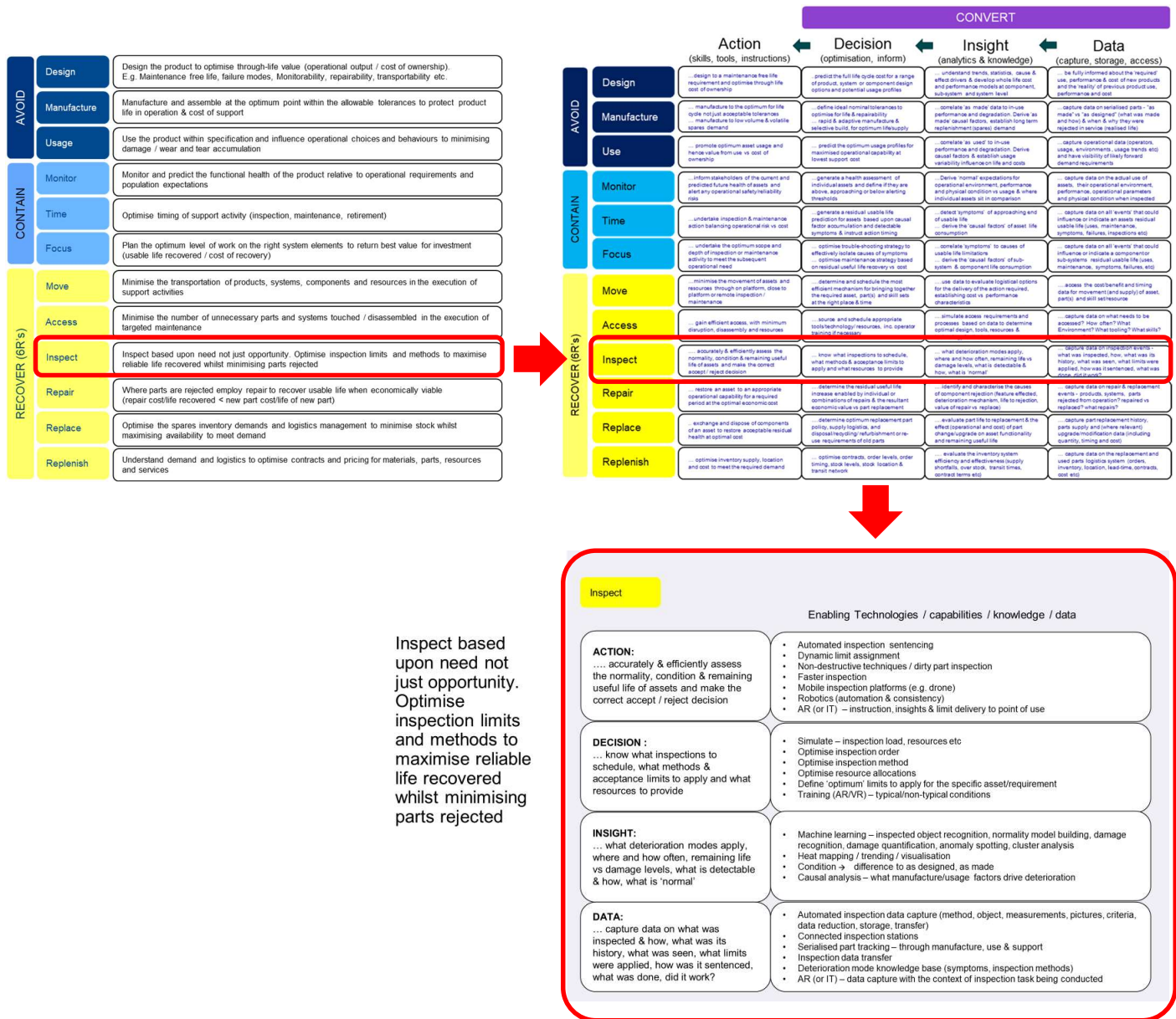
In line with the TES councils focus on the engineering aspects of asset services, we focus in this document on the first two of these capability groups.

¹ See white paper (#1):: *The Value Proposition*

² See white paper (#2): *Service Options*

As illustrated in the following diagram, the body of this paper describes a cascade of capability descriptions from high level representation of the major service elements through to the differentiating lower-level data, knowledge, decision making and physical activity capabilities that enable them.

These descriptions were created by a collective of TES council members representing leading industrial TES providers, leading TES focused academic teams and the Innovate UK Catapult network.



Whilst not exhaustive, the breakdown highlights many of the key area of capability that are perceived to differentiate higher value Through-Life services and offer competitive advantages to both their providers and customers. The breakdown also provides a potential guide to service capability creators and consumers, providing at least a partial checklist against which to evaluate the need for enabling capabilities, their current level of development and future areas of development opportunity.

Content

Discussion

What activities does a Through-Life service need to deliver?

Through-life engineering services predominantly focus on the ability to extract the maximum value from assets in the context of an ever changing usage environment, age based asset deterioration and usage based deterioration.

These change factors lead to an inevitable outcome where the asset is either not being used in the context for which it was designed, or it has deteriorated to a level where it is no longer able to function as required.

The core of Through-Life engineering services therefore exists to avoid, contain or recover from the potential for assets to fail to perform as required throughout their useful lifespan, and to maximise the total net value they deliver over that lifespan.

White paper #2 Service Options discussed a matrix of deliverables that the various themes of service offering may provide.

		Value stream			
		Avoid	Contain	Recover	Convert
Service Level	Value (value/cost optimisation)	Life cycle cost optimised design Asset financing/leasing Value/cost optimised guidance: - Usage - Maintenance timing, scope - Disposal & residual value recovery	Cost surveillance & opportunity alerting	Capability & life Upgrades Inspect/Repair innovations	Value/cost analytics, modelling & optimised decision support
	Availability (right assets, right place, right time, right quantity)	Availability optimised design (lifespan, up-time & maintainability) Availability risk optimised scheduling (inspection, maintenance, inventory, resources) Asset Leasing services Logistics planning	Spare assets, Lease pools & supply services Issue & campaign management	Transportation & storage Support infrastructure, tooling, resources, reactive maintenance planning	Availability risk analytics, modelling & optimised decision support
	Capability (healthy & functional assets)	Capable, robust design Smart assets (sensing, analysing & communicating) Risk optimised maintenance scheduling Inspection instructions Inspection schedule Inspection service Asset Health monitoring	Asset health alerting Fault diagnostics & troubleshooting Technical support	Maintenance instructions Spare parts supply Repair & Overhaul capabilities Remove & install service	Usage, health, failure risk and residual useful life analytics, modelling & optimised decision support
	Usability (Safe, usable assets)	Technical instructions for use Safety upgrades	Technical support Safety risk alerting	Safety issue management (investigate & resolve)	Safety analytics, modelling & optimised decision support

So, the question becomes: what activities lead to these deliverables?

Various resources are available to provide a framework for examining the activities that make up Through-Life engineering services.

Amongst them are:

- The British Standards Institute publication PAS280 - Through-life engineering services – Adding business value through a common framework – Guide (available via the BSI website)
- The Institute of Asset Management – an anatomy (available via the IAM website)

We can identify common features from these frameworks:

- Business Context:
 - Establishing a business commercial framework, including the setting of strategy and policy
- Risk and Issues:
 - Understanding and managing risk and its causes/impacts,
 - Management of emergent issues
- Planning:
 - Predicting load and capacity and adapting plans to manage variance and change
- Decision making:
 - The ability to convert data and information into the decisions about what action to take and when.
- Data:
 - Acquiring and managing the data required to enable effective decision making
- Asset life cycles:
 - Enacting the activities that have the most value at each stage of an assets life cycle from conceptual design through to retirement from operation and management of residual risks
- Organisations and people:
 - Creating and operating the full supply chains required to deliver the service

For the purpose of this discussion we will simplify these common elements into 3 themes:

1. **Physical:** The set of physical activities performed throughout the life cycle of an asset that create it, maintain it and retire it from operation.
2. **Decisions:** The flow of activities from data acquisition through to decision making and instruction that define the content and timing of activity.
3. **Context:** The set of activities that maintain a supply chain and commercial framework that allows the supply chain to operate together to deliver the service.

Physical Activity:

For the purpose of this discussion the set of physical activities below are shown at a high level of abstraction and assume that the service provider is contractually incentivised to maximise the net value from use of an asset (maximising use/output and minimising cost of an engineered product).

The description of each activity focuses on the specific requirements and differentiators of a value incentivised service contract, as opposed to activities that may take place in the absence of such a contractual arrangement. It is these differentiating activities that generate incremental value of a Through-Life engineering service.

AVOID	Design	Design the product to optimise through-life value (operational output / cost of ownership). E.g. Maintenance free life, failure modes, Monitorability, repairability, transportability etc.
	Manufacture	Manufacture and assemble at the optimum point within the allowable tolerances to protect product life in operation & cost of support
	Usage	Use the product within specification and influence operational choices and behaviours to minimising damage / wear and tear accumulation
CONTAIN	Monitor	Monitor and predict the functional health of the product relative to operational requirements and population expectations
	Time	Optimise timing of support activity (inspection, maintenance, retirement)
	Focus	Plan the optimum level of work on the right system elements to return best value for investment (usable life recovered / cost of recovery)
RECOVER (6R's)	Move	Minimise the transportation of products, systems, components and resources in the execution of support activities
	Access	Minimise the number of unnecessary parts and systems touched / disassembled in the execution of targeted maintenance
	Inspect	Inspect based upon need not just opportunity. Optimise inspection limits and methods to maximise reliable life recovered whilst minimising parts rejected
	Repair	Where parts are rejected employ repair to recover usable life when economically viable (repair cost/life recovered < new part cost/life of new part)
	Replace	Optimise the spares inventory demands and logistics management to minimise stock whilst maximising availability to meet demand
	Replenish	Understand demand and logistics to optimise contracts and pricing for materials, parts, resources and services

Decisions:

The process of decision making can be simplified to 3 major steps:

- **DATA:** collecting or acquiring the data necessary to enable decisions to be made.
 - In practice there are 3 sub elements of data acquisition; **capture** of the data at the point where it is available to be created, **storage** of the data in a medium that allows for its subsequent use and **access** of the data by those who have the need for the data and the rights to access and/or use it.
- **INSIGHT:** extracting sufficient meaning and understanding from the data to enable decisions.
 - **Analytics** enable the meaning from new and existing data to be extracted and embedded in **knowledge** stores that codify the learning into re-usable forms such as algorithms, policies, models, training etc.
- **DECISION:** selecting the optimum action to instruct based upon the available insight.
 - **Optimisation** of decisions will be based upon both demand for action and the availability of supply to perform the actions. Decisions may be either **reactive**, as a result of a current evidenced need, or **pro-active** based upon the prediction that without action a need (with potentially higher consequences and hence costs) will occur.

Business Context:

Any Through-Life service is enabled within a business context of stakeholder organisations, individuals and the political, social and commercial frameworks that link them. The normal activities of creating a business strategy, commercial model, supply chain, resources, processes, sales, contracts and operational controls all apply.

Given the complexity of possible contexts in which Through-Life services may be applied we will not labour these elements in this discussion, other than to highlight the importance of through-life thinking as a pervasive theme in the successful implementation of Through-Life services. When a service is predicated on creating true incremental value from use of an asset, all aspects of the business plan, supply chain and contractual model should align to this objective.

What capabilities are required to enable the activities?

Combining the physical activity and decision-making dimensions leads to a more granular breakdown of differentiating Through-Life service delivery capabilities as shown below:

		CONVERT			
		Action (skills, tools, instructions)	Decision (optimisation, inform)	Insight (analytics & knowledge)	Data (capture, storage, access)
AVOID	Design	... design to a maintenance free life requirement and optimise through life cost of ownership	... predict the full life cycle cost for a range of product, system or component design options and potential usage profiles	... understand trends, statistics, cause & effect drivers & develop whole life cost and performance models at component, sub-system and system level	... be fully informed about the 'required' use, performance & cost of new products and the 'reality' of previous product use, performance and cost
	Manufacture	... manufacture to the optimum for life cycle not just acceptable tolerances ... manufacture to low volume & volatile spares demand	... define ideal nominal tolerances to optimise for life & reparability ... rapid & adaptive manufacture & selective build, for optimum life/supply	... correlate 'as made' data to in-use performance and degradation. Derive 'as made' causal factors, establish long term replenishment (spares) demand	... capture data on serialised parts - 'as made' vs 'as designed' (what was made and how) & when & why they were rejected in service (realised life)
	Use	... promote optimum asset usage and hence value from use vs cost of ownership	... predict the optimum usage profiles for maximised operational capability at lowest support cost	... correlate 'as used' to in-use performance and degradation. Derive causal factors & establish usage variability influence on life and costs	... capture operational data (operators, usage, environments, usage trends etc) and have visibility of likely forward demand requirements
CONTAIN	Monitor	... inform stakeholders of the current and predicted future health of assets and alert any operational safety/reliability risks	... generate a health assessment of individual assets and define if they are above, approaching or below alerting thresholds	... Derive 'normal' expectations for operational environment, performance and physical condition vs usage & where individual assets sit in comparison	... capture data on the actual use of assets, their operational environment, performance, operational parameters and physical condition when inspected
	Time	... undertake inspection & maintenance action balancing operational risk vs cost	... generate a residual usable life prediction for assets based upon causal factor accumulation and detectable symptoms & instruct action timing	... detect 'symptoms' of approaching end of usable life ... derive the 'causal factors' of asset life consumption	... capture data on all 'events' that could influence or indicate an assets residual usable life (uses, maintenance, symptoms, failures, inspections etc)
	Focus	... undertake the optimum scope and depth of inspection or maintenance activity to meet the subsequent operational need	... optimise trouble-shooting strategy to effectively isolate causes of symptoms ... optimise maintenance strategy based on residual useful life recovery vs cost	... correlate 'symptoms' to causes of usable life limitations ... derive the 'causal factors' of sub-system & component life consumption	... capture data on all 'events' that could influence or indicate a component or sub-systems residual usable life (uses, maintenance, symptoms, failures, etc)
RECOVER (6R's)	Move	... minimise the movement of assets and resources through on platform, close to platform or remote inspection / maintenance	... determine and schedule the most efficient mechanism for bringing together the required asset, part(s) and skill sets at the right place & time	... use data to evaluate logistical options for the delivery of the action required, establishing cost vs performance characteristics	... access the cost/benefit and timing data for movement (and supply) of asset, part(s) and skill set/resource
	Access	... gain efficient access, with minimum disruption, disassembly and resources	... source and schedule appropriate tools/technology/resources, inc. operator training if necessary	... simulate access requirements and processes based on data to determine optimal design, tools, resources & technology for the task	... capture data on what needs to be accessed? How often? What Environment? What tooling? What skills? What cost? What timescale?
	Inspect	... accurately & efficiently assess the normality, condition & remaining useful life of assets and make the correct accept / reject decision	... know what inspections to schedule, what methods & acceptance limits to apply and what resources to provide	... what deterioration modes apply, where and how often, remaining life vs damage levels, what is detectable & how, what is 'normal'	... capture data on inspection events - what was inspected, how, what was its history, what was seen, what limits were applied, how was it sentenced, what was done, did it work?
	Repair	... restore an asset to an appropriate operational capability for a required period at the optimal economic cost	... determine the residual useful life increase enabled by individual or combinations of repairs & the resultant economic value vs part replacement	... identify and characterise the causes of component rejection (feature affected, deterioration mechanism, life to rejection, value of repair vs replace)	... capture data on repair & replacement events - products, systems, parts rejected from operation? repaired vs replaced? what repairs?
	Replace	... exchange and dispose of components of an asset to restore acceptable residual health at optimal cost	... determine optimum replacement part policy, supply logistics, and disposal/recycling/refurbishment or re-use requirements of old parts	... evaluate part life to replacement & the effect (operational and cost) of part change/upgrade on asset functionality and remaining useful life	... capture part replacement history, parts supply and (where relevant) upgrade/modification data (including quantity, timing and cost)
	Replenish	... optimise inventory supply, location and cost to meet the required demand	... optimise contracts, order levels, order timing, stock levels, stock location & transit network	... evaluate the inventory system efficiency and effectiveness (supply shortfalls, over stock, transit times, contract terms etc)	... capture data on the replacement and used parts logistical system (orders, inventory, location, lead-time, contracts, cost etc)

It is worth noting that for a service to be viable, it requires at least a minimum level of capability for each of the elements of activity and decision making necessary to deliver the service. Higher value services, and service providers, are differentiated by their ability to perform these activities more effectively or more economically than their competitors.

Service providers must therefore aim for a minimum capability level across all the required elements of their offering, whilst developing and exploiting areas of differentiation for selected elements.

It is worth noting that most organisations with significant experience in delivering Through-Life engineering services see the decision-making process (planning the activities to undertake) as the true value creator of the service, with the physical enactment of services being more highly commoditised. The organisation or supply chain that is the most capable at rapidly, accurately and optimally converting operational data into the decisions about what service action to take (or not take) and when, is the one who can best enable incremental value from an assets use at the lowest cost of use.

Appendix 1 below provides a further, but not exhaustive, breakdown which highlights many of the key areas of potential service capability differentiation that are available for exploitation. This breakdown highlights the data streams, knowledge, capabilities and enabling technologies that a collective of TES council members identified as areas of opportunity for current and future differentiation of Through-Life services.

Input Received

This paper has been funded by the InspectorPlus research programme (Bloc Digital, ATI, NATEP, University of Derby, DFS-Consulting).



The content is based upon the combined experience of many industrial and academic organisations, shared through membership and activities sponsored by the TES Council and other collaborative research and industrial peer group activities (including authoring the British Standards Institute guidance **PAS-280 – Through-life Engineering Services** and the creation of the UK national apprentice standard **Through-life engineering specialist - integrated degree - level7**)



Appendix 1 – Differentiation enablers

Design the product to optimise through-life value (operational output / cost of ownership).

E.g. Maintenance free life, failure modes, Monitorability, repairability, transportability etc.

Design

Enabling Technologies / capabilities / knowledge / data

ACTION:

...design to a maintenance free life requirement and optimise through life cost of ownership.

- Design for safety.
- Design for reliability
- Design for maintenance free life.
- Design for inspectability/monitorability.
- Design for accessibility
- Design for repair
- Design for upgrade/update/replacement
- Design for disposal/recycling/reallocation

DECISION:

...predict the full life cycle cost for a range of product, system or component design options and potential usage profiles.

- Operational usage predictions. What is the scope of use that is expected.
- Prediction of through-life attributes. (safety, reliability, maintenance free life...) for a given design and operational usage profile.
- Simulate through-life deterioration and maintenance activities.
- Cost profile estimation for predicted life cycle events.
- Multivariable optimisation. (automation of design based on safety, reliability, maintenance, cost)

INSIGHT:

... understand trends, statistics, cause & effect drivers & develop whole life cost and performance models at component, sub-system and system level

- Historical record, what have we experienced?, how much where and when?
- Classify and cluster to recognise patterns.
- Quantify and rank.
- Context mapping. E.g timelines, geography, people
- Comparitors. Anomaly spotting, trend spotting
- Causality mapping
- Tacit knowledge capture and sharing
- Context comparison, current design vs past operational experience.

DATA:

... be fully informed about the 'required' use, performance & cost of new products and the 'reality' of previous product use, performance and cost

- What is it? What should it be capable of? How was it manufactured? When was it installed? How was it used? What have we done to it (inspected, repaired, replaced)? What did each activity cost? How much was it used? What output did it produce? Did it work as expected? What disruption did it cause? What was it exposed to? Market benchmarking/feedback.
- Sensing, transmission, storage, security, search, accessibility, trading, generation, sharing, cleansing, quality.

Manufacture and assemble at the optimum point within the allowable tolerances to protect product life in operation & cost of support

Manufacture

Enabling Technologies / capabilities / knowledge / data

ACTION:

... manufacture to the optimum for life cycle not just acceptable tolerances
... manufacture to low volume & volatile spares demand

- Automation in the flexibility of production to accommodate changes in demand
- Closed loop control of the machines in the factory
- Adaptive quality/tolerances to the through-life requirements
- Automation of inspection of parts
- Self aware parts - telling the machine what to do
- IoT in the factory (for the parts)
- Automation in producing things for certification
- Targeting optimum tolerance.

DECISION :

...define ideal nominal tolerances to optimise for life & reparability
... rapid & adaptive manufacture & selective build, for optimum life/supply

- Optimising manufacturing processes based on customer demand
- Optimising logistics - trade-off purchase vs make
- Optimising tolerance level for the service life
- Optimised maintenance in the factory to achieve target quality and scale
- Optimising logistics - distributed manufacture.
- Optimising transit/storage/inventory

INSIGHT:

...correlate 'as made' data to in-use performance and degradation.
Derive 'as made' causal factors, establish long term replenishment (spares) demand

- Understanding impact of manufacturing process on part tolerances
- Trade -off tolerances with manufacturing processes
- Life analysis of parts after/during manufacture- remaining useful life
- Decide suitable contract options for different levels of tolerance/quality
- Feedback to design - is the target tolerances achievable in manufacturing?
- As built vs as designed.
- Understanding impact of variation in manufacturing processes on asset life

DATA:

...capture data on serialised parts - "as made" vs "as designed" (what was made and how) & when & why they were rejected in service (realised life)

- Collecting data about manufacturing process, material properties, tracking purchase orders, quality and tolerances, service data as feedback >>>> sensors, smart components, smart machine, monitoring the process, parts movement to the factory/in the factory, data portal for parts storage, inspection of quality. Production monitoring.
- Collecting data for lessons learnt in a structured manner
- Storing the data and distributing it to specific stakeholders
- Specifying data capture requirements.
- Serialised parts tracking and traceability. (back to birth records)

Use the product within specification and influence operational choices and behaviours to minimising damage / wear and tear accumulation

Use

Enabling Technologies / capabilities / knowledge / data

ACTION:

... promote optimum asset usage and hence value from use vs cost of ownership

- In-service product modifications
- Feedback to operators to extend the life
- Feedback to designers on what should be done based on usage profiles
- Condition based maintenance / minimise unscheduled maintenance
- Operator/ customer training

DECISION :

... predict the optimum usage profiles for maximised operational capability at lowest support cost

- Optimisation of usage profiles
- Trade-off operational capability and support cost
- Optimisation of uptime over longer duration of time
- Minimising damage/wear and tear accumulation
- Evaluate if there is a need to modify the plan based on usage profiles

INSIGHT:

...correlate 'as used' to in-use performance and degradation. Derive causal factors & establish usage variability influence on life and costs

- Degradation modelling
- Failure monitoring
- Analyse cause and effect
- Forecasting performance, demand, and reliability
- Analysing predicted vs actual usage and performance

DATA:

...capture operational data (operators, usage, environments, usage trends etc) and have visibility of likely forward demand requirements

- Capture operational data>>> sensor technologies, mapping connectivity, varying environmental conditions, usage profile
- Capture forward demand requirements (forecast vs actual) >>> order book, usage patterns of product, external market factors, reliability

Monitor and predict the functional health of the product relative to operational requirements and population expectations

Monitor

Enabling Technologies / capabilities / knowledge / data

ACTION:

...inform stakeholders of the current and predicted future health of assets and alert any operational safety/reliability risks

- Informatics – communication for effective timely action

DECISION :

...generate a health assessment of individual assets and define if they are above, approaching or below alerting thresholds

- Optimise alerting criteria and thresholds to achieve effective safety and reliability protection with minimum false alerts and waste costs
- Knowledge push & search for 'similar' scenarios and relevant experience to support investigation & decisions

INSIGHT:

...Derive 'normal' expectations for operational environment, performance and physical condition vs usage & where individual assets sit in comparison

- Machine learning – inspected object recognition, normality model building, damage recognition, damage quantification, anomaly spotting, cluster analysis, optimising use of user manual, maintenance performance
- Heat mapping / trending / visualisation
- Condition analytics - difference to as designed, as made
- Causal analysis – what manufacture/usage factors drive deterioration
- Health prognostic modelling – deterministic/probabilistic deterioration predictions

DATA:

... capture data on the actual use of assets, their operational environment, performance, operational parameters and physical condition when inspected

- Automated inspection and maintenance data capture (method, object, measurements, pictures, criteria, data reduction, storage, transfer)
- Connected inspection and maintenance stations
- Serialised part tracking – through manufacture, use & support
- Inspection and maintenance data transfer
- Deterioration mode knowledge base (symptoms, inspection methods)
- Semantic analysis – event data extraction from textual sources (cause, effect, context)

Optimise timing of support activity (inspection, maintenance, retirement)

Time	Enabling Technologies / capabilities / knowledge / data
ACTION: ...undertake inspection & maintenance action balancing operational risk vs cost	- Automated inspection sentencing - Dynamic limit assignment - Non-destructive techniques / dirty part inspection - Faster inspection and maintenance - Mobile inspection platforms (e.g. drone) - Robotics (automation & consistency) for inspection and maintenance - AR (or IT) – mapping images to deterioration database
DECISION : ...generate a residual usable life prediction for assets based upon causal factor accumulation and detectable symptoms & instruct action timing	- Simulate – inspection and maintenance timing, resources etc - Optimise inspection and maintenance - Optimise inspection and maintenance method - Optimise resource allocations - Define 'optimum' limits to apply for the specific asset/requirement - Training (AR/VR) – typical/non-typical conditions at different times
INSIGHT: ...detect 'symptoms' of approaching end of usable life ... derive the 'causal factors' of asset life consumption	- Machine learning – inspected object recognition, normality model building, damage recognition, damage quantification, anomaly spotting, cluster analysis, optimising use of user manual, maintenance performance - Heat mapping / trending / visualisation - Condition analytics - difference to as designed, as made - Causal analysis – what manufacture/usage factors drive deterioration
DATA: ... capture data on all 'events' that could influence or indicate an assets residual usable life (uses, maintenance, symptoms, failures, inspections etc)	- Automated inspection and maintenance data capture (method, object, measurements, pictures, criteria, data reduction, storage, transfer) - Connected inspection and maintenance stations - Serialised part tracking – through manufacture, use & support - Inspection and maintenance data transfer - Deterioration mode knowledge base (symptoms, inspection methods) - AR (or IT) – data capture with the context of inspection and maintenance task being conducted

Plan the optimum level of work on the right system elements to return best value for investment (usable life recovered / cost of recovery)

Focus	Enabling Technologies / capabilities / knowledge / data
ACTION: ... undertake the optimum scope and depth of inspection or maintenance activity to meet the subsequent operational need	- Automated inspection sentencing - Dynamic limit assignment - Non-destructive techniques / dirty part inspection - Faster inspection and maintenance - Mobile inspection platforms (e.g. drone) - Robotics (automation & consistency) for inspection and maintenance - AR (or IT) – instruction, insights & limit delivery to point of use
DECISION : ... optimise trouble-shooting strategy to isolate causes of symptoms ... optimise maintenance strategy based on residual useful life recovery vs cost	- Simulate – inspection load, maintenance depth, resources etc - Optimise inspection and maintenance depth - Optimise inspection and maintenance method - Optimise resource allocations - Define 'optimum' limits to apply for the specific asset/requirement - Training (AR/VR) – typical/non-typical conditions
INSIGHT: ...correlate 'symptoms' to causes of usable life limitations ... derive the 'causal factors' of sub-system & component life consumption	- Machine learning – inspected object recognition, normality model building, damage recognition, damage quantification, anomaly spotting, cluster analysis, optimising use of user manual, maintenance performance - Heat mapping / trending / visualisation - Condition analytics - difference to as designed, as made - Causal analysis – what manufacture/usage factors drive deterioration
DATA: ... capture data on all 'events' that could influence or indicate a component or sub-systems residual usable life (uses, maintenance, symptoms, failures, etc)	- Automated inspection and maintenance data capture (method, object, measurements, pictures, criteria, data reduction, storage, transfer) - Connected inspection and maintenance stations - Serialised part tracking – through manufacture, use & support - Inspection and maintenance data transfer - Deterioration mode knowledge base (symptoms, inspection methods) - AR (or IT) – data capture with the context of inspection and maintenance task being conducted

Minimise the transportation of products, systems, components and resources in the execution of support activities

Move	Enabling Technologies / capabilities / knowledge / data
ACTION: ...minimise the movement of assets and resources through on platform, close to platform or remote inspection / maintenance	• Supplier/contractor communication to enable better understanding of parts requirements • 3D printing at the point of need (or at a convenient place) • Self-healing • Self-awareness • Localised manufacturing in the context of distributed supply chains
DECISION : ...determine and schedule the most efficient mechanism for bringing together asset, part(s) and skill set	• Scheduling software for parts movements • Optimise a distributed supply network • Optimise transportation in existing network • Optimising the timing of having parts ready to distribute • Aligning the in-service use with service resource availability (balancing supply and demand)
INSIGHT: ...use data to evaluate logistical options for the delivery of the action required	• Data analytics/simulation software • Modelling equipment usage profile • Spare parts availability modelling for the in-service product • Process understanding to evaluate how long tasks take
DATA: ...access the cost/benefit and timing data for movement (and supply) of asset, part(s) and skill set/resource	• Communication, external databases, data mining, internal historic data, global economic data....? • Condition of parts, location of parts, movement of parts, plans for usage of asset,, capability/capacity/availability of suppliers, costs of transportation

Minimise the number of unnecessary parts and systems touched / disassembled in the execution of targeted maintenance

Access	Enabling Technologies / capabilities / knowledge / data
ACTION: ...gain efficient access, with minimum disruption, disassembly and resources	• Selecting the optimum route to access the item of interest with minimal disturbance to the rest of the asset. • Enabling inspection in-situ • Minimise tooling for access • Minimise timescale for access • Minimizing maintainer training requirements. • Instruct disassembly/access
DECISION : ...source and schedule appropriate tools/tech./resources, inc. operator training if necessary	• Maintenance schedule optimisation – ◦ Optimum combination of maintenance activities. ◦ Minimise Impact on asset/other functions ◦ Resources, equipment and people • Automation technologies where human access would require shutdowns (H&S etc.) • Disassembly plan. • Hazard assessments
INSIGHT: ...simulate access requirements and process based on data to determine optimal tools/ resources/tech. for the task	• Simulation of task – software and the right data, process modelling • Catalogue of technologies for remote access (Drone, robot, snake arm etc.) • Human vs AI control? • Optimising disassembly trees (knowing the minimum cost route to access) • Risk assessment. • Provision of optimised technical instructions e.g. AR/VR
DATA: ...understand what I will need to access? How often? Geometry? Environment? Operator/ technology capability	• Design specification, maintenance instructions and schedule – extract from PLM? • Service/maintenance environment – can it be changed/controlled/improved? • Technology options for improving task effectiveness/efficiency • Practical constraints. E.g. maintainer capability, automation capability. • Historical knowledge assembly & search (Actual lead times. Actual resource requirements. Frequency of access need and consequences of access route taken. Operating schedule info. – acceptable down time, impact on asset as a whole)

Inspect based upon need not just opportunity. Optimise inspection limits and methods to maximise reliable life recovered whilst minimising parts rejected

Inspect	
Enabling Technologies / capabilities / knowledge / data	
ACTION: accurately & efficiently assess the normality, condition & remaining useful life of assets and make the correct accept / reject decision	- Automated inspection - sentencing - Dynamic limit assignment - Non-destructive techniques / dirty part inspection - Faster inspection - Mobile inspection platforms (e.g. drone) - Robotics (automation & consistency) - AR (or IT) – instruction, insights & limit delivery to point of use
DECISION : ... know what inspections to schedule, what methods & acceptance limits to apply and what resources to provide	- Simulate – inspection load, resources etc - Optimise inspection order - Optimise inspection method - Optimise resource allocations - Define 'optimum' limits to apply for the specific asset/requirement - Training (AR/VR) – typical/non-typical conditions
INSIGHT: what deterioration modes apply, where and how often, remaining life vs damage levels, what is detectable & how, what is 'normal'	- Machine learning – inspected object recognition, normality model building, damage recognition, damage quantification, anomaly spotting, cluster analysis - Heat mapping / trending / visualisation - Condition → difference to as designed, as made - Causal analysis – what manufacture/usage factors drive deterioration
DATA: ... capture data on what was inspected & how, what was its history, what was seen, what limits were applied, how was it sentenced, what was done, did it work?	- Automated inspection data capture (method, object, measurements, pictures, criteria, data reduction, storage, transfer) - Connected inspection stations - Serialized part tracking – through manufacture, use & support - Inspection data transfer - Deterioration mode knowledge base (symptoms, inspection methods) - AR (or IT) – data capture with the context of inspection task being conducted

Where parts are rejected employ repair to recover usable life when economically viable (repair cost/life recovered < new cost/life recovered)

Repair	
Enabling Technologies / capabilities / knowledge / data	
ACTION: ...restore an asset to an appropriate operational capability for a required period at the optimal economic cost	- Train people effectively to make repair efficient (VR) - Make information accessible to assist the repair(AR) - Verification of repair (AI, Sensing) - Automation of repair (self-healing, robots)
DECISION :determine the residual useful life increase enabled by individual or combinations of repairs & the resultant economic value vs part replacement	- Optimise repair timing, scope, and location (for current intervention) - Optimise the life of the repaired part (dynamic scheduling, condition based) - Optimise the capacity for resources required to deliver repair - Optimise
INSIGHT:identify and characterise the causes of component rejection (feature effected, deterioration mechanism, life to rejection, value of repair vs replace)	- Evaluate whether to repair vs replace (outsource vs remake) - Analysing what state you would get to post-repair - Analyse what needs to be repaired (e.g. what's the scope?)
DATA: ... capture data on repair & replacement events - products, systems, parts rejected from operation? repaired vs replaced? what repairs?	Current health data, beyond economic repair, availability of resources to do repair (people, tools, facilities), obsolescence, history of repair, drawings, user manuals, technical documents,

Optimise the spares inventory demands and logistics management to minimise stock whilst maximising availability to meet demand

Replace

Enabling Technologies / capabilities / knowledge / data

ACTION:

... exchange and dispose of components of an asset to restore acceptable residual health at optimal cost

- Optimise selection of replacement part. (upgrade, replace identical, replace equivalent or replace with used part)
- Optimise disposal, recycle, upcycle, repurpose, reallocate.
- Capture reason for replacement.
- Validate residual life recovered.
- Record asset configuration change.
- Set logistics supply demand

DECISION :

...determine optimum replacement part policy, supply logistics, and disposal/recycling/ refurbishment or re-use requirements of old parts

- Predict optimal replacement part demand signal.
- Parts allocation optimisation.
- Parts disposal route optimisation.

INSIGHT:

...evaluate part life to replacement & the effect (operational and cost) of part change/upgrade on asset functionality and remaining useful life

- Component rejection statistics. (historic replacement part demand)
- Upgrade campaigns and or replacement policy.
- Stock availability. (new and alternative and used parts)
- Cost to replace vs asset life recovered.
- Disposal route value and cost.

DATA:

...capture part replacement history, parts supply and (where relevant) upgrade/modification data (including quantity, timing and cost)

- Maintenance plans
- Inspection records
- historic rates of replacement
- Logistics holdings
- Repairability and economic viability of repair vs recycle
- Allowable configurations of the asset.
- Replacement policies.

Understand demand and logistics to optimise contracts and pricing for materials, parts, resources and services

Replenish

Enabling Technologies / capabilities / knowledge / data

ACTION:

... optimise inventory supply, location and cost to meet the required demand

- Order stock
- Allocate stock
- Re position stock
- Dispose of stock

DECISION :

... optimise contracts, order levels, order timing, stock levels, stock location & transit network

- Predict future stock levels
- Optimising replenishment orders
- Optimising storage locations and quantities.
- Optimising allocation of assets
- Optimum transportation plan

INSIGHT:

.... evaluate the inventory system efficiency and effectiveness (supply shortfalls, over stock, transit times, contract terms etc)

- Demand profiles and volatilities
- Supply profiles and volatilities
- Stock and disposition
- Stock turn shelf time vs shelf life
- Fill rates
- SLA compliance (contractual terms)
- Transportation efficiency.

DATA:

... capture data on the replacement and used parts logistics system (orders, inventory, location, lead-time, contracts, cost etc)

- Contracts and service level agreements
- Lead times to replenish
- Cost to replenish
- Storage costs
- Recycling costs/revenues
- Stock capacity
- Shelf life
- Transit routes, times and costs